

SID:1155212799

Short questions (25%)

Question1

1. You are working for data engineering in the Hong Kong Innovation and Technology Commission. The Hong Kong government is planning to develop some AI models for internal use. Your boss asks you something about Ethical Artificial Intelligence Framework:

a. What department drafts the Ethical Artificial Intelligence Framework? (1%)

The Digital Policy Office (DPO) issued the framework on July 2024

b. What are the four related laws or regulations relevant to harmful artificial intelligence practices? (4%)

1. Mainland Cybersecurity Law

2. Data Security Law

3. Personal Information Protection Law

4. Interim Measures for the Management of Generative AI Services

c. The immigration department uses a facial recognition system to detect human faces in the express E-channel service. What is the level of risk of this model? Explain your choice. (2%)

Unacceptable Risks, E-channel service is biometric identification/recognition and would collect Facial recognition databases

d. The transport department uses a car recognition system to monitor the traffic on the road. What is the level of risk of this model? Explain your choice. (2%)

High Risk, which is AI systems in cars, which is about Critical infrastructure systems. AI may misjudgment many special cases in traffic.

e. Suppose that the Education Bureau trains an AI teacher using generative AI. This model assists the students in solving high school mathematics problems. What is the level of risk of this model? Explain your choice. (2%)

High Risk, as it is AI systems used in education. AI teacher can directly impact on learning assessment and educational outcomes.

f. Suppose that the Hospital Authority trains an AI assistant using deep learning. This model acts as the triage system to set the priority categories of the patients in the Accident and Emergency (A&E). What is the level of risk of this model? Explain your choice. (2%)

High Risk, as it is AI systems in medical devices, which is about Critical infrastructure systems. AI cannot be responsible for any accidents.

g. After studying the examples from part c to part f, suggest three more examples on the use of AI with different departments in the Hong Kong Government. In addition, state and explain the level of risk of your examples. (9%)

1. Education bureaus can use AI to design educational and entertaining games, AI enabled video games is Minimal Risks: (Code of Conducts) as use AI making a game would not cause any danger.

2. Food and Environmental Hygiene Department can use AI to filter different restaurant hygiene complaints, Spam filter and is Minimal Risks: (Code of Conducts) as use AI to filter text complaint would not cause any danger.

3. Leisure and Cultural Services Department can have a footfall forecasting and queue-time estimation, which only involves in data analysis which is Minimal Risks: (Code of Conducts).

h. In the Ethical AI Principles, what are transparency and Interpretability? Why are transparency and Interpretability important? How can transparency and Interpretability be achieved? (3%)

1. What are they

Transparency: Clearly show the way AI works, include the data it uses, its purpose, limits, and assumptions.

Interpretability: Provide explanation which is easy to understand, explain why outputs came by the inputs.

2. Why they are important

It can build the trust and accountability, supports audits and compliance, reduces bias and unfair outcomes and enables safer deployment and improvement.

3. How to achieve

using simpler, models which on earth can be explained, keeping the document and logs, providing the way to appeal and override, monitoring performance for continuously trained model